

Kylie Savoye

PhD Student

 Birmingham, UK

 kms306@student.bham.ac.uk

 +44 7515319870

 [linkedin.com/in/kylie-savoye](https://www.linkedin.com/in/kylie-savoye)

 <https://orcid.org/0009-0008-0111-8541>

EDUCATION

2023 - Present	PhD in Applied Mathematics - Mathematics and Data Analysis for Spatial Transcriptomics University of Birmingham EPSRC & AstraZeneca Scholarship Developing topological data analysis tools for disease and treatment prediction
2022 - 2023	MSc Mathematical Modelling and Scientific Computing St Hugh's College, University of Oxford Dissertation project: "Characterising the solution space of vascular blood flow"
2019 - 2022	BSc Astrophysics (Honours) University College London Bachelor's Research Group project: "Ice, Ice Maybe: A photometric investigation of active processes on the moons of Uranus"

SKILLS

Programming
Python (proficient): data structures, scientific computing, algorithm design, probabilistic modelling
MATLAB (advanced): numerical methods, simulations, data analysis
High-performance/large data workflows

Mathematics
Topological data analysis
Geometry-based modelling
Probability/statistics for data analysis

PUBLICATIONS

June 2026	<u>CLEC14A correlates with neutrophil infiltration in hepatocellular carcinoma and mediates neutrophil recruitment across liver endothelial cells</u> - Journal of Pathology Authors : J.M. O'Rourke, D. Kavanagh, A. O'Keeffe, K. Savoye , K. Yang, D.A. Patten, C.E. Willoughby, J.M. Llovet, G. Neag, H.L. Reeves, P.W. Hewett, N. Kalia, T. Shah, Y. Ting Ma, K. Hunter, J. Flint, J.L. McMurray, O. Cain, F. Spill, D.A. Mann, A.J. Naylor, V. Heath, C.J. Weston, R. Bicknell, S. Shetty Applied computational and spatial analysis methods to investigate cellular organisation and biologically relevant tissue interactions.
Nov. 2025	<u>Spatial transcriptomics links hepatocyte-macrophage interactions to viral signatures in seronegative hepatitis</u> - Preprint Authors: A. Bozward, M. Buhruznia, J. Cole, S. Davies, K. Savoye , C. Das, M. Pugh, G. Wootton, P. Reis Nisa, F. Spill, O. Cain, A. Croft, A. McNally and Y. Oo Analysed spatial transcriptomics data to identify disease-associated cellular interactions and spatial signatures linked to liver pathology.
Sept. 2025	<u>Nano-org, a functional resource for single-molecule localisation microscopy data</u> - Nature Communications Authors: S. Shirgill, D.J. Nieves, J.A. Pike, M.A. Ahmed, H. Abbott, M.H.H. Baragilly, K. Savoye , J.D. Worboys, K.S. Hazime, E. Bruggeman, A. Garcia, D.J. Williamson, P. Rubin-Delanchy, R. Peters, D.M. Davis, R. Henriques, S.F. Lee and D.M. Owen Contributed statistical methodology and simulation studies for quantitative analysis of large-scale super-resolution microscopy datasets
August 2025	<u>Measuring the Similarity of SMLM-Derived Marked Point-Clouds</u> - Biophysical Journal Authors: K. Savoye , D. J. Nieves, S. Shirgill, A. Lewis, F. Spill, and D. M. Owen Developed novel mathematical and computational methods for comparing complex biological imaging datasets and quantifying spatial organisation

ACTIVITIES

Outreach

- Member of CDT in Topological Design Outreach team (2025 – present) aimed to teach school students about topology and its wide range of applications in STEM
- Member of Piscopia Committee (2024 – present), an organisation aimed at encouraging women and underrepresented genders in mathematics in pursue a PhD

Sports

- Birmingham University **W1 Fencing Captain** (2023 - 2025)
 - 5th place at BUCS Individual National Championships (2025)
 - 2024-25 BUCS National Trophy Champions (2025)
 - Promotion to BUCS Premiere North League (2025)
- Oxford University Fencing Club Member of Blues and Assassins team
 - Winner of BUCS Premiere South League (2023)
- University College London **W1 Fencing Captain** (2020-2022)
- Fencing Cadet World Championships (2017)

FURTHER EXPERIENCE

2023 - Present	Post-Graduate Teaching Assistant University of Birmingham Modules: Mathematical Workshops (MATLAB), Numerical Programming (MATLAB and Python) and Applied Mathematical Analysis (Dynamical Systems and Asymptotic methods)
June 2023	Examination Scribe and Reader University of Oxford Provided scribing and reading services to support students with disabilities during examinations
June – Sept. 2022	Summer research Intern in Planetary Systems University College London – Awarded Funding by research group Built a Python computational model to track planet growth and formation using Markov Chain Monte Carlo algorithms to recover probability distributions of the initial properties of a planet
June – August 2021	Summer Research Intern in Galaxy and Quasar Physics, DESI survey University College London – Awarded Brian Duff Studentship Built a Python spectral stacking tool to enable galaxy evolution work within the DESI survey, and wrote a project proposal and funding application

CONFERENCES

July 2026	European Conference on Mathematical & Theoretical Biology – University of Graz, AUT Mini Symposium Talk: Quantifying Tissue Architecture in the Tumour Microenvironment with Multi-Parameter Persistent Homology
May 2026	Foundations of Computational Geometry and Topology – Brown University, USA Poster: Multiparameter Persistent Homology for Spatial Transcriptomics
July 2025	British Early Career Mathematician's Colloquium - University of Birmingham, UK Organising Committee
June 2025	British Applied Mathematics Colloquium - University of Exeter, UK Poster: Persistent Homology in Spatial Transcriptomics: Bridging Spatial and Expression Landscapes
July 2024	International Spatial Biology Congress - The Hague, NL Poster: Measuring the Similarity of SMLM-Derived Marked Point-Clouds
June 2024	European Light Microscopy Initiative - Liverpool, UK Poster and Early Career Researcher Flash Talk: Measuring the Similarity of SMLM-Derived Marked Point-Clouds
Sept. 2023	Piform 2023 - University of Birmingham, UK Poster and Contributed Talk: Characterising the Solution Space of Vascular Blood Flow